Larger Pore Size Hollow Fiber Membranes as a Solution to the Product Retention Issue in Filtration-Based Perfusion Bioreactors

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Larger pore size hollow fiber membranes as a solution to the product retention issue in filtration-based perfusion bioreactors†

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Abbreviations: ATF, alternating tangential flow; **TFF**, tangential flow filtration; **kDa**, kilo-Daltons; **PES**, polyethersulfone; **PS**, polysulfone; **PE**, polyethylene; **kcps,** kilo counts per second

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Abstract

Tangential flow filtration (TFF) and alternating tangential flow (ATF) filtration technologies using hollow fiber membranes are commonly utilized in perfusion cell culture for the production of monoclonal antibodies; however, product retention remains a known and common problem with these systems. To address this issue, commercially available hollow fibers ranging from several hundred kilo-Daltons (kDa) to 0.65 μm in nominal pore size were tested and were all demonstrated to undergo moderate to severe product retention. Further investigation revealed accumulation of particles in the same size range (approximately 20-200 nm) as the pores. Based on the assumption that these particles contribute to product retention and membrane plugging, a hollow fiber with an unconventionally larger pore size was subsequently identified and demonstrated to drastically reduce product retention with no impact to cell clarification, Furthermore, these hollow fibers demonstrated surprisingly high membrane capacities, making them an attractive solution to the problem of product retention in perfusion reactors.

1 Introduction

Filtration devices are a popular option for perfusion cell culture due to their scalability, simplicity, and promise of total cell clarification even at high cell densities [1, 2]. The relative ease of operation, however, is offset by the unpredictable phenomenon of product retention and membrane fouling, which limits the lifetime of such processes and can make commercialization problematic. To such an end, much effort has been made to understand and improve the product retention phenomenon. Operating the filter in TFF mode compared with dead end filtration helps reduce but does not fully alleviate product retention and membrane fouling [3]. Similarly, ATF technologies may offer a further reduction of product retention but do not present a complete solution [4]. Product retention is such a persistent issue that some perfusion processes bypass the issue completely by switching to ultrafiltration mode, intentionally concentrating product within the reactor over the entire duration of the cell culture [5]. This solution, however, does not work for highly labile products, which are traditional candidates for continuous cell culture [6], nor is the batch nature of the harvest conducive for a fully integrated continuous manufacturing process. Another workaround to product sieving is to switch to a new hollow fiber unit when sieving has decayed to a predetermined point; however, product sieving is still present in this solution and the onset of sieving is rapid and severe with a new membrane in our hands. A complete solution to product retention therefore still remains to be discovered.

We have shown that 20nm to 200nm particles are associated with product retention on a 0.2 μm membrane [18]. We hypothesize that rather than using a 0.2 μm filter, a larger pore filter would allow the 20-200nm particles to pass through the membrane, and therefore reduce the amount of product retention.

Several mechanisms of membrane fouling and protein retention have been proposed over the years and include but are not limited to protein deposition [7, 8], surface blockage [9], and cake layer formation [10, 11]. Furthermore, severe membrane fouling was reported even when the pore size of the filter was much larger than the protein of interest [12-15]; however the ranges of pore size tested in most cases did not exceed that of 0.65μm based on the commercial availability of such hollow fibers, which can be of a similar order of size to protein aggregates and other cellular debris. In two notable exceptions, Mercille et al. and De la Broise et al. described the rapid onset of fouling with low porosity track-etched Nuclepore track 5μm and 10μm pore flat-plate filter, respectively. The low product passage could have been influenced by sub-optimal cell viabilities ((17% and 44%), as well as the low porosity [16, 17]. The question still remains if improved product passage can be gained using larger pore sized hollow fibers for lower molecular weight compounds such as antibodies for high viability cell cultures.

To prove the hypothesis, hollow fibers with a wide range of pore sizes were tested. Traditional pore sizes utilized in filtration-based perfusion technologies all experienced product retention; however, switching to hollow fibers with a pore size greater than 2μm allowed for complete product passage in a perfusion process with high cell viability. Unlike previous reports, the membrane capacity of these hollow fibers was surprisingly high (4,000-20,000 L/m2) and, more strikingly, product passage remained at close to 100% for the entire duration of the membrane lifetime. Use of a larger pore size hollow fiber is thus a feasible solution to the product retention issue in perfusion cultures.

2 Materials and methods

2.1 Perfusion cell culture and tangential flow filters

The cell retention device used in all perfusion reactors was a hollow fiber operated in tangential flow filtration (TFF) mode with a magnetically levitated centrifugal pump as the recirculation device (Levitronix, Zurich, Switzerland). Targeted shear through the hollow fiber was kept at 1800 s^{-1} and matched for all experiments. All lab-scale studies were conducted in 3L glass stirred tank reactors at a 2L working volume (Applikon Biotechnology, Delft, Netherlands) with a perfusion volume of 4L/day. Large-scale studies were conducted in a Hyclone 100L single use bioreactor with a perfusion volume of 200L/day. In-house proprietary recombinant Chinese Hamster Ovary (CHO) cell lines producing monoclonal antibodies were used. The cell densities were 30-40 million cells per mL at steady state.

The standard hollow fiber used was a polyethersulfone (PES) hollow fiber with a 0.2µm pore size and 1mm lumen ID (Spectrum, Rancho Dominguez, CA). Polysulfone (PS) hollow fibers with a lumen ID of 1mm and pore sizes of 500kD, 0.1µm, 0.2µm, and 0.45µm (GE Healthcare, Marlborough, MA) were used in offline experiments of product sieving at 2.7L/m^2/hr. 5 µm and 0.2 µm alpha alumina Membralox microfiltration hollow fibers with a 7mm lumen ID and a total surface area of 50cm^2 were purchased from Pall Exekia (Bazet, FR) and used for a 2L perfusion bioreactor at 2.8mL/min or 33L/m^2/hr. The Membralox was tested in a 100L bioreactor with a 0.72m^2 4 mm lumen, 57 channel, 1000 mm long filter operated at 140mL/min or 12. L/m^2/hr.

A prototype Tangential Flow Depth Filters (TFDF) 2-10µm polyethylene (PE) hollow fibers with a 7mm lumen ID, and a membrane thickness of over 2 mm and a total surface area of 50cm^2 was purchased from Spectrum (Rancho Dominguez, CA) and operate at 2.8mL/min or 33L/m^2/hr. A commercially available TFDF KROSFLO was tested at the 100L scale (length, 65cm, pore size 2-5um, material PE, lumen diameter 3mm, 43 fibers, area 2765 cm^2, Spectrum) at 140mL/min or 30. L/m^2/hr.

The experimental set up, tubing lengths and dimensions were the same for all 2L bioreactor experiments.

2.2 Offline model of product retention

A perfusion reactor was harvested at day 8 of cell culture and the harvest centrifuged at 1200rpm for 15 minutes to separate cells and larger debris from supernatant. This supernatant had previously been determined to contribute to product retention and demonstrated good capacity to mimic the phenomenon offline [18]. The supernatant was then cycled individually through new hollow fibers of the specified pore sizes. Both perfusion (permeate flow) and recirculation (reactor flow) were fed back into the original sample reservoir to conserve volume and minimize changes to sample over time. Flow rates were controlled by peristaltic pumps and velocities were set to match shear in the perfusion systems. Samples of reactor and permeate were taken at the indicated time points and IgG concentration measured. Instantaneous product passage is expressed as the ratio concentration of IgG in permeate to concentration of IgG in reactor. 100% product passage indicates no product retention across the membrane.

2.3 Analytical methods

Viability and cell density were measured by Vi-CELL Cell Counter (Beckman Coulter Life Sciences, Indianapolis, IN). Titer was measured using a Cedex BioHT (Roche Diagnostics GmbH, Mannheim, Germany). These titer measurements had been previously verified to be in agreement with in house HPLC methods. Particle size was assessed using a Malvern Zetasizer Nano ZS (Malvern, Westborough, MA). Derived count rate is expressed as kilo counts per second (kcps) and is used as a measure of relative number of total particles within the sample.

3 Results

3.1 Commonly used hollow fibers all experience product retention despite differences in pore size

Standard hollow fibers used for perfusion tend to range in pore size from several hundred kD to several hundred nm. To determine the product retention profiles within this range, cell culture supernatant was run offline through hollow fibers with 500kD, 0.1µm, 0.2µm and 0.45µm nominal pore sizes. Interestingly, the 0.2um hollow fiber offered the best product passage at 5 and 10 minutes after the start of perfusion. Hollow fibers with a smaller pore size saw worse performance compared to the 0.2µm membrane across the entire time course investigated while a 0.45µm hollow fiber had a similar product passage profile with the exception of the two previously mentioned time points (Figure 1A). A 0.2µm hollow fiber was then determined to be the best performing option out of the range tested and was used as the standard for comparison for all subsequent experiments. Figure 1B shows the product retention profile of a typical perfusion run utilizing a 0.2µm hollow fiber.

3.2 Switching to a larger pore size hollow fiber alleviates product retention

Two hollow fibers with nominal pore sizes above 2µm were subsequently identified for comparison: a 5µm hollow fiber made from alpha alumina (Pall) and a prototype 2-10µm hollow fiber made from PE (Spectrum). The prototype filter has a relatively thick (2mm) filter thickness, and is therefore referred to as a "Tangential Flow Depth Filter" (TFDF). These filters were operated at different fluxes than the smaller pore size due to the large lumen diameter. The 7mm lumen diameter precluded matching the shear rate and the membrane area of the different pore size systems while keeping the volumetric flow rate in the lumen below one bioreactor volume every two minutes. We chose to match the shear rate; the membrane area was smaller and thus the flux was higher for the large pore hollow fibers.

Surprisingly, both large pore filters offered nearly complete product passage for the entire lifetime of the filter. Figure 2 shows performance of the two large pore size hollow fibers compared to a 0.2µm alpha alumina and a standard 0.2µm PES hollow fiber. While product passage in a TFF perfusion reactor using the 0.2µm hollow fibers decreases over time, the same process using a larger pore size hollow fiber revealed no change in product passage for over 30 days. These results also suggest that pore size rather than hollow fiber material or hollow fiber lumen diameter makes the most impact on product retention. Due to commercial availability constraints of the hollow fibers, the 0.2µm alpha alumina hollow fiber had considerably less surface area compared to the other small pore filters, partially accounting for the rapid decrease in product passage observed with this filter. Though the 0.2µm alpha alumina hollow fiber plugged within two days, the total volumetric challenge was 1600 L/m^2, which is similar to the other small pore membranes. Furthermore, particle analysis revealed marked accumulation of 20-200nm sized particles within a perfusion reactor using the 0.2µm hollow fibers while the particle numbers stayed consistent between the reactor and permeate stream of a >2µm hollow fiber, suggesting that the larger pore size allows for complete passage of these particles (Figure 2B). Despite the increase in particle load to the harvest stream, initial investigations showed no adverse effects to the downstream purification process (data not shown).

3.3 Product sieving remains at close to 100% for the entire filter lifetime

Because cell debris and other materials might rapidly plug the larger pore hollow fibers resulting in a short filter lifetime, filter capacity was determined by operating a 5µm alpha alumina hollow fiber in TFF perfusion mode until complete plugging was observed. Filter capacity was determined to be cell line specific and was surprisingly large for both cell lines tested (4601 L/m^2 Cell line 1, 8570 L/m^2 Cell line 2). Furthermore, product passage remained at close to 100% up until the end of the membrane lifetime at which time onset of filter plugging was rapid and occurred over the course of several hours (Figure 3). This kind of 'all or nothing' product passage characteristic is more beneficial in terms of process predictability compared to the gradual decrease observed with traditional 0.2µm filters.

The capacity of the TFDF was even larger. The system did not foul in our 2L experiments, and in one case achieved a capacity of 20,000L/m^2 before the experiment was terminated due to media limitations.

The large pore filters were used at the 100L perfusion scale successfully. The alpha alumina filter achieved a capacity in excess of 3,000L/m^2 with nearly 100% product transmission (Figure 4). Due to product sizing limiations, the alpha alumina filter had to be operated at 11L/m^2/hr rather than the 33L/m^2/hr operation at the 2L scale. The PE TFDF filter achieved a capacity of 11,00L/m^2 with nearly 100% product transmission. The cell densities for both runs were between 30 and 40 million cells per mL , and the viabilities remained above 85% for the duration of the run (data not shown).

4 Discussion

Product sieving and membrane fouling are known issues with filtration-based perfusion systems. Here, we offer a solution for the product sieving phenomenon by utilizing hollow fibers with a drastically large (>2µm) pore size. These hollow fibers offered complete cell retention while allowing for complete passage of product. More impressively, the filter capacity of these hollow fibers was surprisingly large, facilitating scale up. While filter plugging was inevitable, product passage remained at close to 100% for the entire duration of the filter lifetime.

To the best of the authors' knowledge, this is the first report of utilizing larger pore sizes to solve the product sieving issue in perfusion reactors. While larger pore sized filters (5 and 10µm flat plate) in perfusion-based cultures had been previously reported [16, 17], these filters fouled quickly and required change out every four to five days. Indeed, an argument was put forth against the necessity of going to such large sizes as no obvious benefit could be discerned [2]; however, these studies were complicated by low cell culture viabilities. As all the perfusion processes described in this paper maintained high cell viabilities (>90%), which facilitated direct loading onto a protein A column, what this process would look like at different viabilities has yet to be determined. Conceivably, there might be some viability cut-off where membrane capacity is negatively impacted and use of such large pore size hollow fibers is no longer feasible. The amount of capacity reduction caused by such low viability could severely impact run duration or necessitate an excessively large filter surface area, making operation and scale up difficult. Alternatively, the large pore filters used here have not just filter area, but filter depth. Both the PE and alumina filters have an active filtration area that potentially uses the pores inside the membrane matrix to provide additional capacity.

The quality of the material being purified downstream would also need to be reassessed. Particle size analysis already revealed the presence of a greater amount of larger sized particles in the permeate stream, which may increase with greater levels of cell death. The impact to the purification process when considering a fully integrated continuous process would need to be considered when switching to a hollow fiber with a larger pore size.

Conflict of interest

The authors declare no financial or commercial conflict of interest.

5 References

[1] Voisard, D., Meuwly, F., Ruffieux, P. A., Baer, G., Kadouri, A., Potential of cell retention techniques for large-scale high-density perfusion culture of suspended mammalian cells. *Biotechnol. Bioeng.* 2003, 82, 751-765.

[2] Woodside, S. M., Bowen, B. D., Piret, J. M., Mammalian cell retention devices for stirred perfusion bioreactors. *Cytotechnology* 1998, 28, 163-175.

[3] van Reis, R., Shishir, G., Frautschy, L. N., Orlando, S., et al., High Performance Tangential Flow Filtration. *Biotechnol. Bioeng.* 1997, 56, 71-82.

[4] Clincke, M. F., Molleryd, C., Zhang, Y., Lindskog, E., et al., Very High Density of CHO Cells in Perfusion by ATF or TFF in WAVE Bioreactor - Part I. Effect of the Cell Density on the Process. *Biotechnol. Prog.* 2013, 29, 754-767.

[5] Yang, W. C., Minkler, D. F., Kshirsagar, R., Ryll, T., Huang, Y. M., Concentrated fed-batch cell culture increases manufacturing capacity without additional volumetric capacity. *J. Biotech.* 2016, 217, 1-11.

[6] Pollock, J., Ho, S. V., Farid, S. S., Fed-batch and perfusion culture processes: economic, environmental, and operational feasibility under uncertainty. *Biotechnol. Bioeng.* 2013, 110, 206-219.

[7] Matthiasson, E., The role of macromolecular adsorption in fouling of ultrafiltration membranes. *J. Membrane Sci.* 1983, 16, 23-36.

[8] Clark, W. M., Bansal, A., Sontakke, M., Ma Y. H., Protein adsorption and fouling in ceramic ultrafiltration membranes. *J. Membrane Sci.* 1991, 55, 21-38.

[9] Attia, H., Bennasar, M., de la Fuente, T. B., Study of the fouling of inorganic membranes by acidified milks using scanning electron microscopy and electrophoresis. *Journal of Dairy Research* 1988, 58, 51-65.

[10] Belfort, G., Davis, R. H., Zydney, A. J., The Behavior of Suspensions and Macromolecular Solutions in Crossflow Microfiltration. *J. Membrane Sci.* 1994, 96, 1-58.

[11] Kelly, S.T., Zydney, A.L., Protein fouling during microfiltration: comparative behavior of different model proteins. *Biotechnol. Bioeng.* 1997, 55, 91-100.

[12] Baruah, G. L., Belfort, G., A predictive aggregate transport model for microfiltration of combined macromolecular solutions and poly-disperse suspensions: model development. *Biotechnol. Prog.* 2003, 19, 1524-1532.

[13] Bowen, W. R., Gan, Q.. Properties of microfiltration membranes: flux loss during constant pressure permeation of bovine serum albumin. *Biotechnol. Bioeng*. 1991, 38, 688-696.

[14] Herrero, C., Pradanos, P., Calvo, J. I., Tejerina, F., Hernandez, A., Flux Decline in Protein Microfiltration: Influence of Operative Parameters. *Journal of Colloid and Interface Science* 1997, 187, 344-351.

[15] Kelly, S. T., Zydney, A. L., Effects of intermolecular thiol-disulfide interchange reactions on BSA fouling during microfiltration. *Biotechnol. Bioeng.* 1994, 44, 972-982.

[16] Mercille, S., Johnson, M., Lemieux, R., Massie, B., Filtration-based perfusion of hybridoma cultures in protein-free medium: Reduction of membrane fouling by medium supplementation with DNase I. *Biotechnol. Bioeng.* 1994, 43, 833-846.

[17] De la Broise, D., Noiseux, M., Massie, B., Lemieux, R., Hybridoma perfusion systems: A comparison study. *Biotechnol. Bioeng.* 1992, 40, 25-32.

[18] Wang, S., Godfrey, S., Ravikrishnan, J., Lin, et al., s-1 contributions to cell culture performance and product recovery in ATF and TFF perfusion systems. *J. Biotechnol.* 2017, 246, 52-50.

Figures

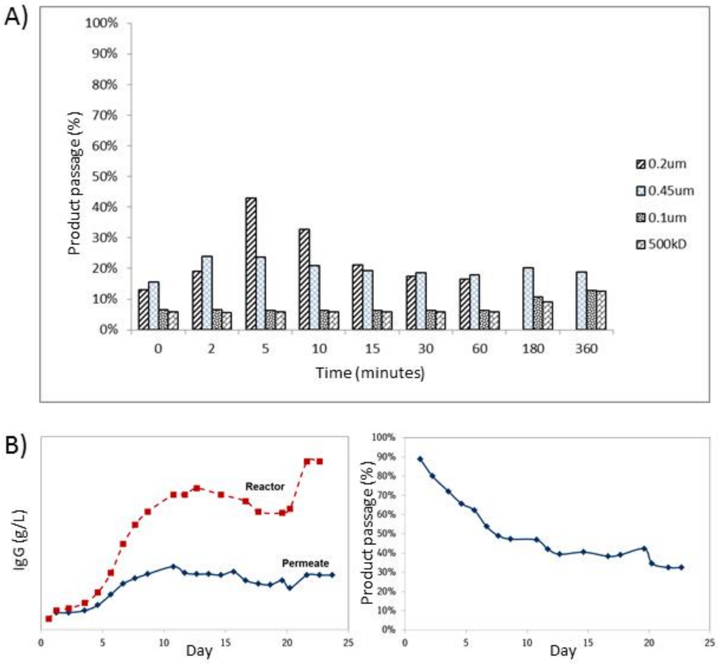

Figure 1. A) Product passage was assessed in hollow fibers with pore sizes of 500 kD, 0.1µm, 0.2 µm, and 0.45 µm in an offline model of product retention. IgG concentration was measured from samples taken from the reactor side and the permeate side at the indicated time points and expressed as percent product passage. B) Product retention profile of a 2L perfusion reactor operated in TFF mode using a 0.2 µm hollow fiber operated at 2.7L/m^2/hr. The total membrane challenge after 24 days of perfusion was 1600L/m^2.

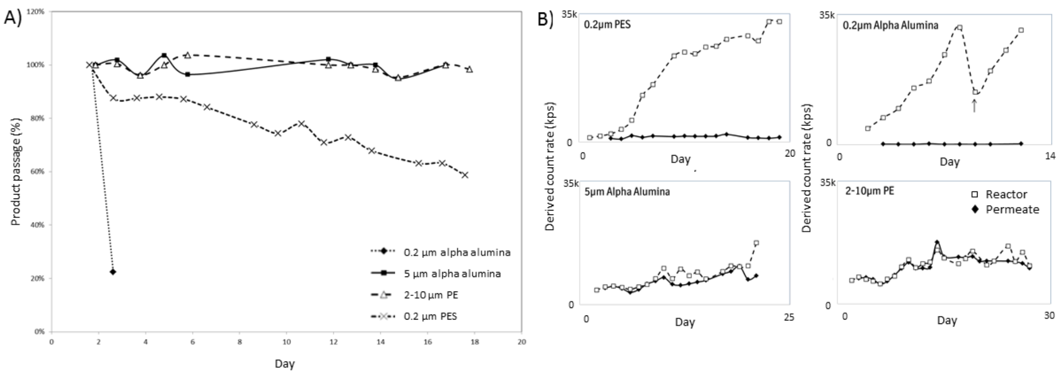

Figure 2. TFF perfusion reactors were run with a permeate flow of 2.8mL/min using four different hollow fibers: 0.2 µm PES operated at 2.7L/m²/hr, 0.2 µm alpha alumina operated at 33L/m²/hr, 5 µm alpha alumina operated at 33L/m²/hr, and 2-10 µm polyethylene PE TFDF operated at 33L/m²/hr. Reactor and permeate samples were taken daily and analyzed for A) product retention and B) particle count. The arrow denotes when the half the total working volume of the reactor was harvested and the total volume brought back up by diluting with fresh media. The total membrane challenge was 1200L/m² for the 0.2 µm PES, 1600L/m² for the 0.2 µm alpha alumina, 14,000L/m² for the 5 µm alpha alumina, and 20,000L/m² for the 2-10 µm polyethylene PE TFDF

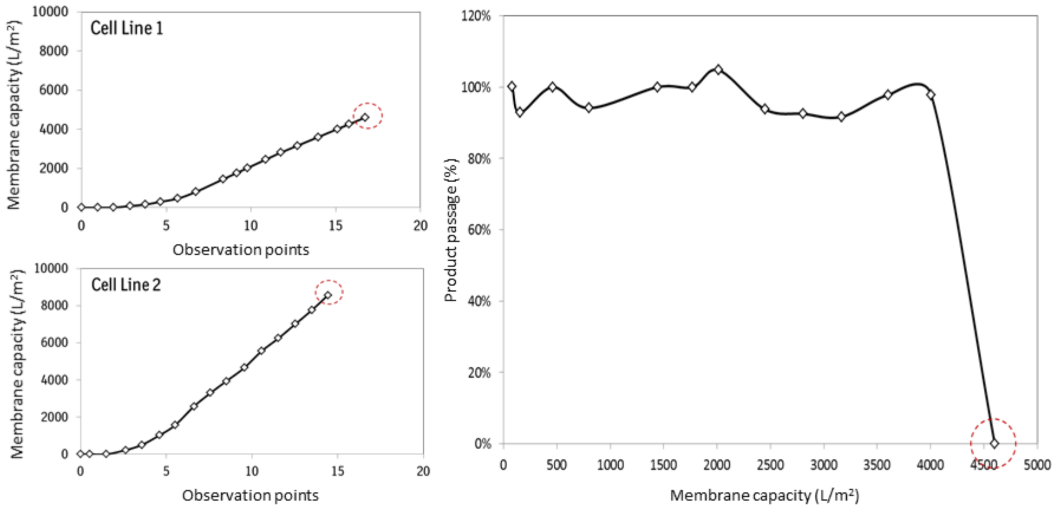

Figure 3. Two different cell lines were operated in TFF perfusion mode using a 5μm alpha alumina hollow fiber to determine membrane capacity. Filter capacity is defined as volume of liquid perfused through the hollow fiber over the filter surface area. Red circles denote time when membrane became completely plugged as denoted by inability to draw any permeate across the filter.

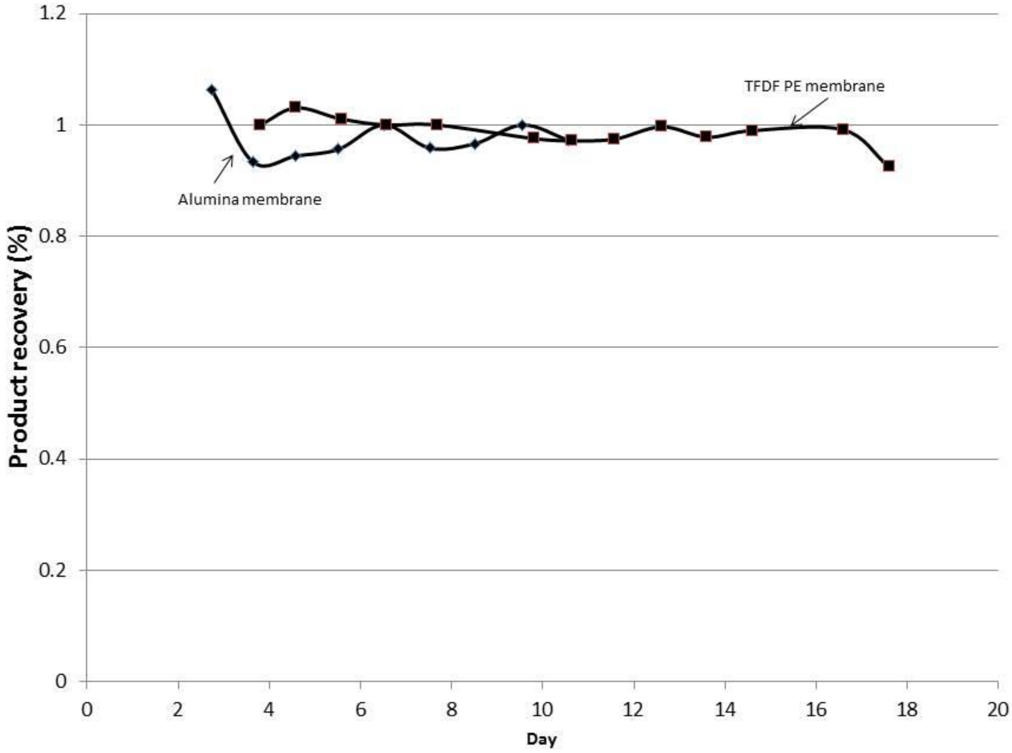

Figure 4. The 5um alpha alumina hollow fiber and the 2-5 um PE TFDF were operated with a 100L single use bioreactor. The capacity of the alumina fiber was in excess of 3,000L/m^2, as an unrelated issue ended the run prematurely. The TFDF was in excess of 11,000L/m^2. Both filters had product transmission levels near 100% for the duration of the run.